HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported manufacturing processes and drove improvements for existing products by analysing production data and implementing corrective actions, reducing minor defect rates by **15%**.
- Defined and validated manufacturing processes for upgrade products, collaborating with design engineers to ensure manufacturability and reducing assembly time by **10%**.
- Provided support for mechanical/electrical assembly jigs, fixtures, and purpose-built test equipment, ensuring operational readiness and minimising downtime.
- Conducted process analysis and Product Quality reporting, translating raw data into actionable insights for engineering and production teams.
- Collaborated with cross-functional teams of Engineers and Production Staff to uphold quality standards and identify areas for enhancement in manufacturing workflows.
- Assisted Product Designers by providing feedback on product designs to ensure ease of assembly and efficient production, contributing to a **5%** reduction in assembly rework.
- Containerised key manufacturing support tools and deployed them using **Docker** on **AWS EC2** instances to improve deployment consistency and manageability.

SKILLS

Manufacturing Processes – Process Improvement, Process Definition, Process Validation, Manufacturing Environment, Six Sigma, Lean Projects

Data Analysis & Reporting – Process Analysis, Product Quality Reporting, Excel, Business Intelligence Tools, Statistical Process Control, Gauge Studies, Process Capability

Engineering Tools & Technologies – Assembly Jigs, Fixtures, Test Equipment, Technical Drawings, GD&T;'s, Industry 4.0, Real-time Data Analytics

Programming & Scripting – SQL, VBA, Excel Macros, Python

Cloud & Containerisation – AWS, EC2, S3, Docker

Collaboration & Problem Solving – Structured Problem-Solving, Cross-functional Collaboration, Interpersonal Skills, Organisation Skills

PROJECTS

Manufacturing Process Optimisation for Component Production

- Analysed production data from a mid-volume manufacturing cell using **SQL** and **Excel** to identify bottlenecks and quality deviations.
- Developed and validated new assembly procedures for a key component, incorporating feedback from production staff and reducing assembly errors by **12%**.
- Documented findings and recommendations for process improvements, presenting them to the engineering management team for review and implementation.

Test Equipment Automation for Quality Assurance

- Developed automated test scripts using **Python** and **VBA** to streamline the quality assurance process for manufactured sub-assemblies.
- Integrated test equipment with a custom data logging system, capturing key performance metrics and enabling real-time quality reporting.
- Reduced manual testing time per unit by **30%**, allowing QA engineers to focus on complex defect analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)